

Data and Modelling Infrastructure for Living Labs

Service Working Group



IAT-Partner:

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VERSITÄT

JUSTUS-LIEBIG-
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Hochschule
Geisenheim
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Gefördert durch:

Leibniz
Leibniz-Gemeinschaft

Federal Ministry
of Food
and Agriculture

LAND
BRANDENBURG
Ministry for Science,
Research and Culture

HESSEN

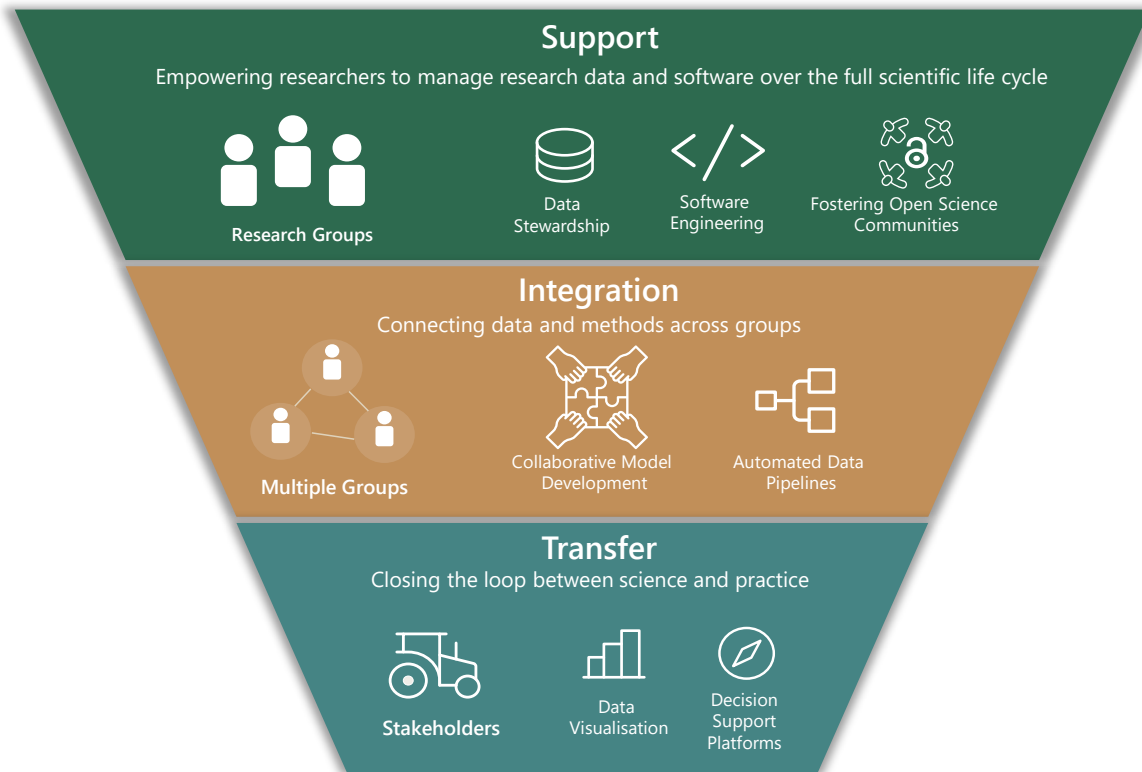
Data and Modelling Infrastructure for Living Labs

WHO WE SERVE

WHAT WE PROVIDE



Ben Black



Data and Modelling Infrastructure for Living Labs

WHO WE SERVE

WHAT WE PROVIDE

Support

Empowering researchers to manage research data and software over the full scientific life cycle



Research Groups



Data
Stewardship



Software
Engineering



Fostering Open Science
Communities

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Empowering researchers to manage research data and software over the full scientific life cycle



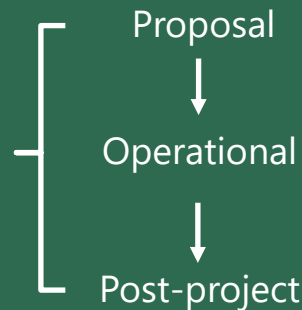
Research Groups



Data
Stewardship



Software
Engineering



Improving FAIRness

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Research Groups



Data
Stewardship



Software
Engineering



Fostering Open Science
Communities



Case by case support
+
Needs-based training



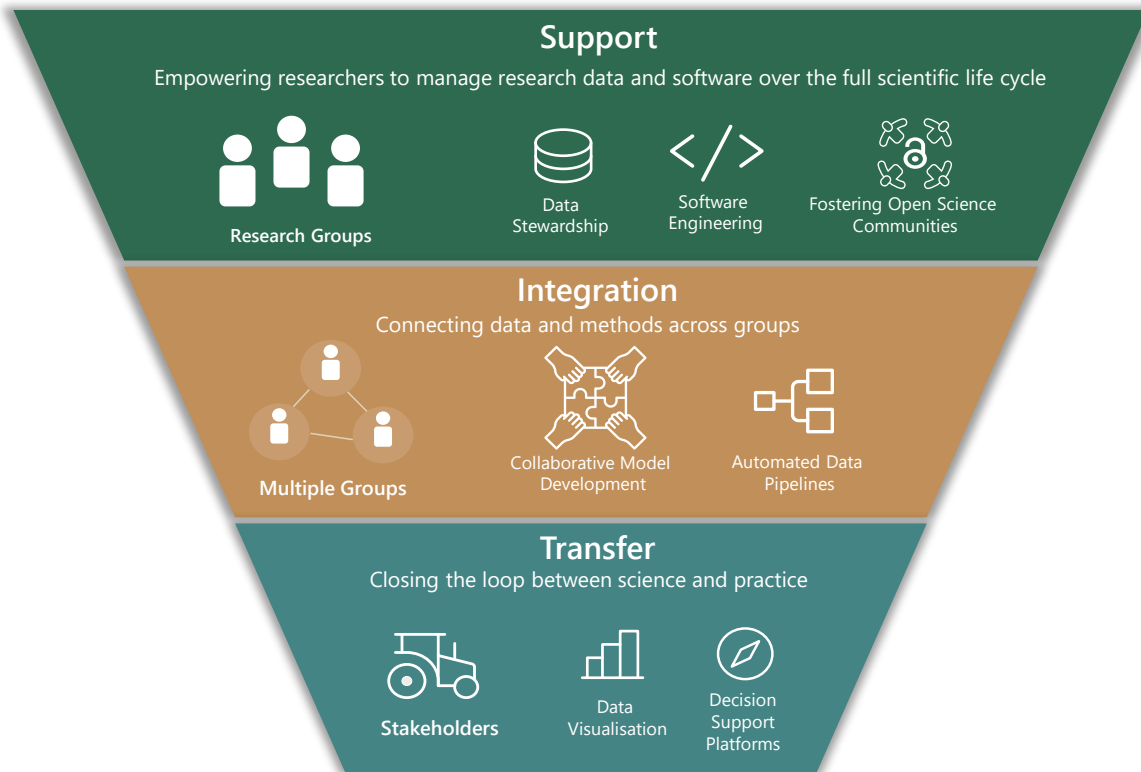
In
collaboration
with:

CDP

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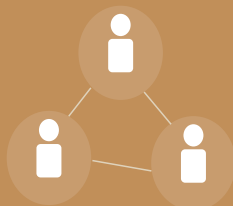
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WHO WE SERVE

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Integration

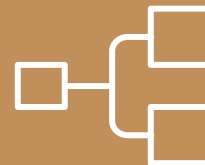
Connecting data and methods across groups



Multiple Groups



Collaborative Model
Development



Automated Data
Pipelines

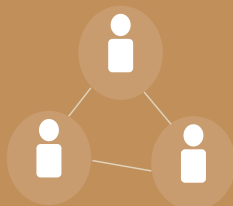
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Integration

Connecting data and methods across groups



Multiple Groups



Collaborative Model
Development

Assistances + Resources



Participatory modelling
and scenario
development



Model Integration

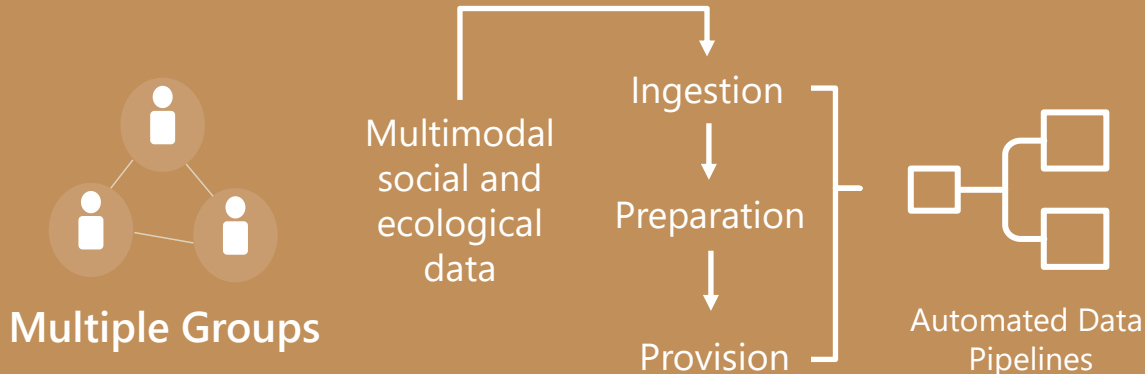
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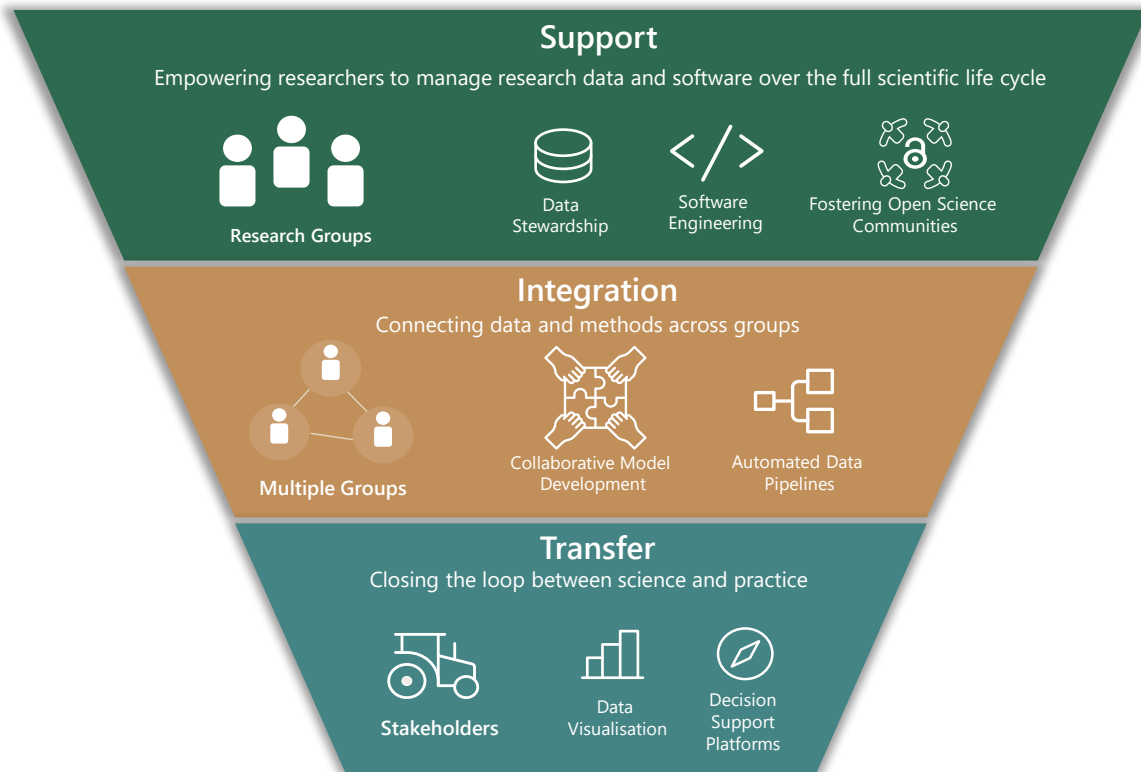
Connecting data and methods across groups



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Transfer

Closing the loop between science and practice



Stakeholders

User-centred



Dashboards



Interactive
applications



'Story-telling'
with data



Data
Visualisation



Decision Support
Platforms



Data and Modelling Infrastructure for Living Labs

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Build-It-Yourself: Low-Cost Systems for Field Ecophysiology An Open Handbook for DIY Environmental Measurement Systems

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Contributors

Researchers:

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This handbook is part of the MonksHillLab initiative (integral part of the Working Group Ecophysiology of Water and Matter Cycling of the Leibniz Centre for Agricultural Landscape Research) which aims to democratize science by enabling low-cost, DIY sensor development for environmental monitoring. It provides step-by-step guidance on building and calibrating sensor systems using open hardware and widely available components.

The project supports ecophysiological research and the discovery of sustainable agricultural management practices, particularly in under-resourced regions and the Global South. By focusing on reproducibility, accessibility, and practical application, it seeks to close the gap between research/development and field implementation/adoption.

This is a living document: users are encouraged to build upon the existing designs, share adaptations, and contribute new chapters. You're very welcome to share your improvements/chapters. Just email the author (mathias.hoffmann@zalf.de) and submit an update. Future versions will include additional low-cost DIY systems and their integration with the MonksHillLab Logger App. Thus, together, we aim to grow with that handbook a practical, evolving resource for scientists, students, and practitioners.

We are currently exploring the idea of organizing a low-cost sensor summer school and/or online workshop to provide hands-on training, exchange ideas, and build a global community around DIY environmental monitoring. If you are interested in participating or helping shape the format, feel free to get in touch or respond to the feedback link: <https://docs.google.com/forms/d/e/1FAIpQL56-IZMhYAK-It-eMoKwW0KqrdHeFuPRd5Pp-JMQW3SEVip/viewform?usp=header>

Whats New? Version 2.1 introduces the EVE-Offline and EVE-Online workflows including a backend infrastructure and dashboard setup for real-time data transfer and visualization (EVE-Online). In addition Appendix D was added, now containing supporting data starting with data for the PAR sensor validation as well as the proof of concept of EVE-Offline and EVE-Online.

Whats Next? Version 2.2 will add a new chapter presenting a low cost system for flush-sampling of chamber headspace air for lateron analyses at a GCMS, as well as the introduction of a new regional hub located in Germany.

Files

Improved Accessibility and Reuseability

1 Hardware Solutions

1.1 Environmental Variables

1.1.1 NiMH Solar Trickle Charger

1.1.1.1 Purpose & Use Case:

The solar trickle charger module presented here is designed to provide a simple, low cost, and field-deployable solution for recharging AA/NiMH batteries in remote areas. Its primary use cases is to support small-scale electronic devices—such as environmental sensors or data loggers—used in off-grid research and monitoring setups. Especially relevant in the Global South and other regions where access to stable electricity is limited, this charger enables reliable, low-maintenance battery recharging using solar energy. By leveraging basic components like a small solar panel and a Schottky diode, the system offers a practical way to maintain battery-powered devices in the field without the need for complex charging infrastructure or frequent battery replacement.

1.1.1.2 Bill of Materials:

A comprehensive overview of all components required to build the solar charger, needed quantities, recommended suppliers and approximate prices (in €) is provided in [Table 1.1](#).

Table 1.1: Components required for solar charger assembly (includes quantities, typical suppliers, and approximate prices (not listed per item due to frequent changes)).

Component	Amount	Supplier	Price (approx.)
Schottky Diode	1	Amazon, Reichelt, Conrad	0.10€
Common E-Series electronic resistor (e.g. 500Ω)	1	Amazon, Reichelt, Conrad	0.10€
AAA Battery holder	1	Amazon	3.00€
Perfordard / MonksHillLab PCB board (2x2 cm)	1	Amazon	0.50€
Wire (Black, red) / Screw Terminal Blocks	-	Amazon	0.50€
6V 2.5W Solar Panel	1	Amazon	10.00€
Sum			14.20€

1.1.1.3 Wiring Diagram:

A schematic overview of the wiring layout, showing all necessary connections between electrical components, is provided in [Figure 1.1](#).

Figure 1.1: Wiring scheme for the PAR sensor. Illustrating the electrical connections between the photodiode, resistor, and microcontroller. The schematic focuses solely on the wiring layout required for sensor assembly, not placement of additional, non-electrical components in the field.

VIDE



Data
Visualisation

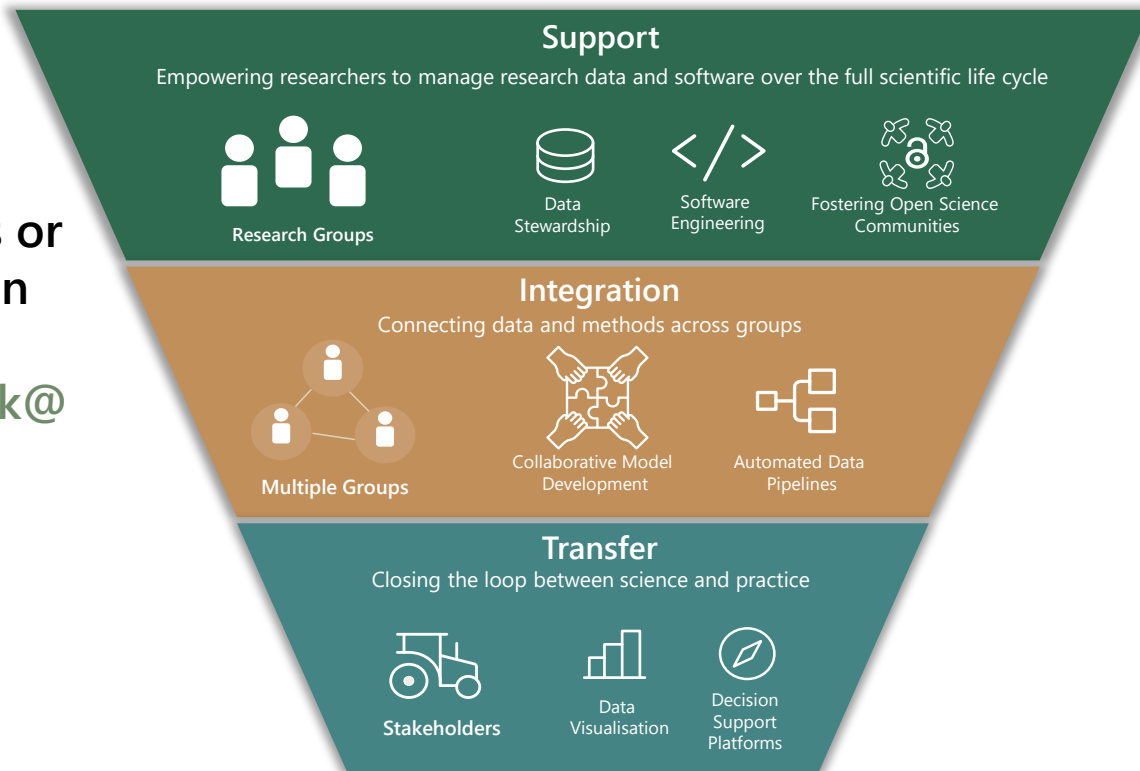
Platforms

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If you have questions or ideas for collaboration please reach out:
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